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Professional Development Increases Child Care Providers' NDBI Implementation and Children's Language Outcomes

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Abstract

A majority of children's language learning experiences occur in inclusive early child care and education settings. Few evidence-based professional development (PD) programs exist to empower early childhood education providers to use language instruction practices with children in inclusive classrooms. There is little research on providers' implementation of naturalistic developmental behavioral intervention (NDBI) strategies to improve children's language outcomes in whole-group inclusive instruction. We conducted a pilot cluster randomized controlled trial to test the hypothesis that providers in the Early Achievements for Child Care Providers (EA-CP) PD program would show greater gains in NDBI implementation fidelity during shared book reading relative to business-as-usual (BAU) providers. This hypothesis was confirmed. Compared to BAU providers, EA-CP providers exhibited significantly greater fidelity gains from pre-training to mid-training, post-training, and 12-week maintenance assessments. A secondary hypothesis was that toddlers with and without developmental delays in the EA-CP condition would exhibit greater vocabulary and communication gains than toddlers in the BAU condition. Analyses of children with and without delays showed significantly greater expressive vocabulary and story-related engagement and communication in the EA-CP group. For children with delays, positive effects favoring the EA-CP group but not statistically significant were found. Implications for research, practice, and policy are discussed.

Keywords

professional development, components of practice, child care, language and communication, child development, disabilities and development delays

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Early childhood care and education (ECCE) providers (hereafter, providers) play a vital role in children's development of language (Piasta, 2016) and pre-academic skills (Meloy et al., 2019); their instructional practices have long-term effects on language and academic performance (National Early Literacy Panel, 2008). Of the 59% of children age 5 years and younger enrolled in at least one weekly non-parental care arrangement, 62% attend a day care center (Cui & Natzke, 2021). Many (about 14%) have an *identified* learning-related developmental delay or disorder (DD) (e.g., learning disability, attention-deficit hyperactivity disorder, autism) (Boyle et al., 2011). Even more (25%) are at elevated risk for developmental, behavioral, or social delay (National Center for Health Statistics at the Centers for Disease Control, 2013). Per the Americans with Disabilities Act (1990), licensed child care centers must provide care for children with DD. However, gaps persist that limit full, meaningful access and inclusion. Two major gaps involve availability of professional development (PD) (a) for early childhood education and care providers to support implementation of evidence-informed child language-promoting instruction in inclusive classrooms and (b) studies examining outcomes of both providers and children (Hindman & Wasik, 2023). Using preliminary data from a two-phase iterative development process, we designed and assessed a PD program to address these gaps via a pilot randomized controlled trial (RCT), preliminarily assessing PD effects on providers' fidelity and children's outcomes. Results will inform researchers and policy makers about core PD ingredients, including PD dosage, needed to ignite fidelity gains in early childhood providers in inclusive classrooms, and implications for advancing children's language skills.

Vocabulary-Focused PD for Inclusive Early Childhood Providers

Language skills, especially vocabulary, represent an essential instructional target in early childhood education settings because of their critical importance for children's long-term academic success (Young et al., 2002), particularly in the domain of reading (Dickinson et al., 2010; National Early Literacy Panel, 2008; Quinn et al., 2015). Despite the importance of language development during early childhood, early childhood providers' instructional practices do not provide frequent, enriched language learning opportunities (Wright & Neuman, 2013). Rather than offering rich child language production experiences (important language stimulation strategy, Barnes et al., 2017), teachers produce 93% of the talking in early childhood classrooms (Dickinson & Caswell, 2007), most of which is directive (e.g., giving instructions for cleaning up) (Barnes et al., 2017). Indeed, only about 5 min per day is spent engaging children in conversational exchange or explicitly teaching vocabulary, with no increase during book sharing (Deshmukh et al., 2019).

For children with language learning challenges, the lack of language enrichment is particularly noteworthy. When addressing children with language delay, providers use significantly more directive language and less language-modeling input than when addressing children with no delays (Girolametto, Weitzman, et al., 2000). Hindman and colleagues (2019) reported that, during book reading, teachers rarely used strategies such as open-ended prompting and making book-related comments to stimulate verbal participation and language formulation from children at risk for DD. Rather, teachers predominantly used closed prompts, such as yes/no or "test" questions for information already established in children's knowledge and skills, and rarely paused for child reflection and commenting (Girolametto, Hoaken, et al., 2000; Hindman et al., 2019). When teachers did ask open-ended questions, they usually accepted an answer from only one child per question, resulting in abbreviated information exchanges and language processing opportunities for children. These instructional practices inhibit children's participation and are associated with lower vocabulary learning (Hindman et al., 2019). Not surprisingly, teachers report lack of training to instruct children with DD (Vesay, 2008) as well as lower self-efficacy for working with children with disabilities (Guo et al., 2021). Strategically designed PD programs are needed to effectively teach providers effective language facilitation strategies and support their high-fidelity instruction to promote children's language learning.

We designed a PD program based on a logic model with four key ingredients for improving children's language (emphasizing vocabulary) development. The ingredients consisted of (a) a multi-modal instructional approach (e.g., including gestures and props), and (b) improving providers' implementation of naturalistic developmental behavioral intervention strategies (NDBI; Schreibman et al., 2015) in (c) a familiar instructional context rife with language-promoting instructional opportunities (shared book reading) by (d) augmenting workshop-provided explicit implementation guidance with job-embedded, individualized explicit coaching. Below, the rationale for each of these ingredients is presented.

Multi-Modal Vocabulary Instruction Integrating Gestures and Props

Our approach to vocabulary instruction was multi-modal, involving the use of gestures and props. Given its close relation to vocabulary learning (Iverson & Goldin-Meadow, 2005), pairing gesture with target vocabulary usage has several potential instructional benefits. For example, when teaching the new word "yawn," the spoken word could be paired with a gesture of an open mouth and covering the mouth with one's hand. Doing so could help children expand existing conceptual knowledge and map new linguistic forms onto those concepts (Goldin-Meadow, 2000). Teacher-provided modeling and elicitation of gestures representing target vocabulary may heighten children's attention to target words, provide increased processing time, and foster word learning. In addition to gestures, story-related props facilitate attention engagement and embodied learning (Wasik & Bond, 2001). Props also offer children concrete referents during extratextual discussion using targeted vocabulary. The multi-modal and embodied experience of concepts that occurs through gesture production and story enactment using props afford opportunities to learn vocabulary words' semantic features. This may efficiently enhance new, deep word learning (Rosborough, 2014). Therefore, our PD program taught providers how to incorporate target vocabulary-related gestures and the use of props while implementing NDBI strategies during shared book reading.

NDBI Strategies

The instructional procedures in our intervention, NDBI strategies, integrated evidence-based instructional strategies from applied behavior analysis (e.g., three-part contingencies; prompt hierarchy) and developmental science (e.g., modeling, balanced turns, baiting child initiation, integration into naturalistic and meaningful contexts). NDBI strategies have been successfully implemented in shared book reading by early childhood pre-K teachers and providers; positive effects on children's social communication outcomes resulted (Engelstad et al., 2020; Feuerstein & Landa, 2020). NDBI strategies are well-aligned with key Division of Early Childhood Recommended Practices for Instruction (Division of Early Childhood, 2014, indicated using the abbreviation "INS"), such as teaching within meaningful routines (e.g., book sharing; INS5), providing clear and developmentally appropriate cues (antecedents) for targeted behavior production (e.g., child saying a target vocabulary word), prompting (e.g., pausing, modeling; INS6), and reinforcing child attempts using contextually relevant rewards (INS7).

Most research on NDBI implementation and its effects on children's language development has been conducted in 1:1 clinician-child or caregiver-child interventions with autistic or other DD populations (Kaiser & Roberts, 2013). The only RCTs known to us examining NDBI strategies implemented by providers in community classroom whole-group learning contexts or in a shared book reading context focused on children with autism (Engelstad et al., 2020; Feuerstein & Landa, 2020). To our knowledge, no published RCTs have focused on implementation by providers in community-based inclusive early child care centers and assessed outcomes of providers as well as children with and without DD. To address these gaps, advance the field's ability

to support providers' inclusive practices, increase providers' access to evidence-informed, high-quality community-based care for young children with DD, research to develop and evaluate the efficacy of language-focused PD programs is needed (Walker et al., 2020).

Shared Book Reading as an Implementation Context

Shared book reading, which occurs nearly daily in most child care classrooms, provides an ideal context for implementing NDBI strategies to teach vocabulary and promote children's verbal engagement related to instructional targets. The information conveyed via storybook text and illustrations establishes a meaningful context for vocabulary learning, promoting the formation of semantic networks and deep word learning. Further, illustrations in children's books expose children to novel words and ideas, provide clear referents, and often, provide multiple exemplars for target vocabulary. NDBI strategies can be easily embedded into shared book reading, especially when targeting children's expressive language use (Hargrave & Sénéchal, 2000; Storkel et al., 2019). For example, providers may give clear and developmentally appropriate antecedent cues for the specific behavior sought from children (e.g., "Look! Here is the fireman's hose. He needs to hook it to the _____!"), then deliver prompting strategies if the child does not approximate the targeted behavior. Also, providers would give natural reinforcement (e.g., "You said 'hydrant'! That's right! The fireman hooks the hose to the hydrant!"), model and facilitate balanced turns (e.g., "It's a hydrant! Tell Michael, "Hook the hose to the hydrant!"), and so on.

Coaching Ingredients

Currently, most PD opportunities for early childhood providers involve workshops or courses (Cunningham et al., 2015; Hamre et al., 2012) without explicit coaching support to apply the content to each participant's individualized instructional setting and learning needs. Researchers must address the gap in availability of language-focused PDs that are time-efficient, individualized, and feasibly integrated by busy providers in inclusive child care settings. PD programs also must integrate evidence-based ingredients to have maximal effects on providers' vocabulary enrichment practices during book sharing because changing the instructional practices of busy providers, especially practices involving their linguistic behavior, is difficult (Hindman & Wasik, 2023). Three key PD ingredients have been identified as particularly essential per a meta-synthesis. These PD ingredients were associated with positive educator *and* student outcomes (Dunst et al., 2015): (a) training dosage of at least 15 hr, distributed over time (Diamond et al., 2013); (b) ongoing support through a collaborative job-embedded coaching process involving observation, reflection, explicit feedback and action planning (Snyder et al., 2015), permitting individualization of explicit implementation guidance; and (c) active engagement of teachers during training. Further, the PD should target explicitly defined instructional practices and incorporate guided reflection (Snyder et al., 2015) since providers need specific, concrete guidance on how to implement targeted instructional strategies (Hindman & Wasik, 2023). PD programs that integrated the above components have yielded early childhood educator outcomes of improved knowledge and skills (e.g., Egert et al., 2018).

Present Study: Research Questions

Addressing the existing research gap in RCTs examining PD programs targeting vocabulary instruction for children with and without DD in inclusive early childhood settings (Brunsek et al., 2020), the present pilot RCT was conducted. The primary research questions were as follows: (a) Does the PD program improve providers' fidelity of implementation of NDBI strategies during

shared book sharing? We hypothesized that two interactive workshops paired with 12 weekly synchronous job-embedded coaching sessions would activate improved fidelity of NDBI implementation. (b) Do children in trained providers' classrooms with and without delays demonstrate greater vocabulary and communication outcomes compared to those in untrained providers' classrooms? We hypothesized that children in trained classrooms would make greater gains.

Method

Design

Using implementation science principles (Aarons et al., 2011), the Early Achievements for Child Care Providers (EA-CP) PD program was iteratively developed over 3 years through extensive systematically elicited stakeholder (providers, directors) input and data collection. Results informed the PD program's duration, implementation context, workshop content/number/timing, and delivery of in-person rather than virtual coaching (Pfeiffer et al., 2022; Pfeiffer & Landa, 2022). A cluster RCT design (Puffer et al., 2005) was used. Data on a single cohort of providers and children were collected from baseline to maintenance during the COVID-19 pandemic (fall 2021-spring 2022). All study procedures were approved by the Institutional Review Board at the Johns Hopkins University School of Medicine.

Recruitment

Providers and children were recruited from inclusive child care centers in three urban and suburban counties in a mid-Atlantic state. The study coordinator contacted directors of licensed child care centers on lists provided by a state agency to inform them about the study via email and/or phone. Follow-up conversations were conducted via Zoom with interested directors and eligible providers; study purpose, consent process, incentives, and procedures were discussed. Upon completing the online consent process, a recruitment email template and study flyer were shared with the provider to recruit children and families from the classroom. Interested families completed an online consent process for their child's participation and baseline testing at the child care center was scheduled. Providers' incentives, regardless of group assignment, included: (a) continuing education hours through the State Department of Education Office of Child Care for workshop participation, (b) free books (four for providers randomized to the PD [EA-CP] condition, two for providers randomized to business-as-usual [BAU]) and accompanying props (story-related pictures and objects), and (c) up to $360 in gift cards for completing whole-group book readings. Children's caregivers received a report of their child's assessment results and a $25 gift card incentive for completing electronic developmental screeners.

Participants

Providers. Twenty-six providers were enrolled, 13 per condition. Participating providers were required to: (a) work at a state-licensed center-based inclusive child care center, (b) teach children ages 2 and/or 3 years, (c) teach at least 3 days per week, and (d) have at least one child with developmental delay in their classroom consented to participate in the study. Two providers in the EA-CP condition withdrew from the study during the intervention period (one due to a medical leave of absence and one due to a personal injury). Most providers were female (90.9%) and White (54.2%). Providers' education levels varied: 29.2% had a high school diploma or GED equivalent, 20.8% had an Associate's degree or some college experience, 33.3% had a Bachelor's degree, and 16.7% had a Graduate degree. Providers' teaching experience ranged from 2 to 31 years ($M = 13$, $SD = 9.6$). All EA-CP (11/11) and 11/13 BAU providers reported having state certification.

Children. There were 124 children enrolled at the start of the study (62 per condition), ages 24 to 48 months ($M = 33$, $SD = 6.6$). Eligibility criteria included: (a) being between ages 24 and 48 months at the start of the study, (b) attending the child care center at least 3 days per week, (c) being exposed to English at least 50% of the time at home, and (d) having a hearing device or glasses/contact lenses if a hearing or visual impairment was present. Children with ($n = 59$) and without developmental delays ($n = 65$) were included. Criteria for delay included: (a) scoring in the "monitor" or "further assessment needed" ranges of the Ages and Stages Questionnaire-3 (ASQ-3; sensitivity = .86; specificity = .85; concurrent validity 74% to 100%; Squires & Bricker, 2009); (b) scoring in the "possible delay" or "further assessment needed" range of the Ages and Stages Questionnaire: Social Emotional, Second Edition (ASQ:SE-2; sensitivity = .81; specificity = .83; validity = .84; Squires et al., 2015); or (c) having a previous diagnosis of a DD (e.g., speech, language). Both ASQ screeners are parent-completed questionnaires that provide reliable, accurate developmental and social-emotional screening. During the intervention period, seven children withdrew from the EA-CP condition (six because their provider withdrew; one left their center) and two withdrew from the BAU condition (one moved to a non-participating classroom; one left their center). Demographic and baseline developmental data of the remaining 115 children (EA-CP: $n = 55$, 26 with delays; BAU: $n = 60$, 33 with delays) are presented in Table 1. The Visual Reception subscale of the Mullen Scales of Early Learning (Mullen, 1995) provided participant characterization for nonverbal cognitive development.

Randomization

Intervention assignment occurred at the center level, preventing contamination across conditions. All providers within the same center were randomized to either the EA-CP or BAU control condition. A statistician simulated all possible randomizations using the R statistical program (version 4.0.3) to identify the randomization schema with the best balance across arms. The outcome was one that achieved the following parameters: (a) both conditions had an equal number of providers and a nearly equal number of children, (b) all providers within the same center (maximum of two) were randomized to the same condition, and (c) the proportion of children with DD were balanced between conditions. Among the possible randomization solutions that emerged using these parameters, the final solution was identified as having no statistically significant differences between conditions in terms of both provider (education, gender, race, ethnicity, years of teaching experience) and child variables (parental education, income, gender, race, ethnicity, delay status). Providers assigned to the EA-CP condition received the PD program and implemented the EA-CP instructional approach (detailed below). Providers assigned to the BAU condition conducted instruction as usual and attended workshops (same as those received by the EA-CP group) after post-training data collection. Figures 1 and 2 provide CONSORT flow diagrams (Schulz et al., 2010) for providers and children, respectively, from enrollment through data analysis.

EA-CP Instructional Approach Procedures

The supplemental book-sharing instructional approach taught to providers in the EA-CP PD program (described below) resulted from a translational process across three prior interventions, beginning with a clinician-implemented, center-based, group NDBI for autistic toddlers (Landa et al., 2011). Subsequent translations for different types of community-based early childhood education settings retained the NDBI strategy implementation, but increasingly simplified the instructional elements and contexts to maximize the feasibility of implementation (Feuerstein & Landa, 2020; Pfeiffer & Landa, 2022). Further adaptations for this study were needed due to pandemic influences on the child care climate, including provider exhaustion and concerns about

Table 1. Child Baseline Characteristics for EA-CP and BAU Conditions.

Variable	Total	EA-CP ($n = 55$)		BAU ($n = 60$)		p
		No delay ($n = 29$)	Delay ($n = 26$)	No delay ($n = 27$)	Delay ($n = 33$)	
Child age (mean, *SD*) (months)	33.4 (6.6)	34.5 (6.5)	33.8 (7.3)	33.1 (5.4)	32.2 (7.1)	.21
Child age (range) (months)	24–48	25–47	25–45	25–48	24–48)	
Child sex (percent)						
Female	49.6	44.8	46.2	63.0	45.5	.51
Male	50.4	55.2	53.8	37.0	54.5	
Child race (collapsed; percent)						
Black, non-Hispanic	13.0	10.3	7.7	18.5	15.2	.46
White, non-Hispanic	77.4	82.8	80.8	77.8	69.7	
Other	9.6	6.9	11.5	3.7	15.2	
Parent income (percent)						
≤49k	17.9	23.1	50.0	0.0	0.0	.14
50k–99k	42.9	30.8	50.0	50.0	57.1	
≥100k	39.3	46.2	0.0	50.0	42.9	
Parent education (percent)						
High school grad or some college	3.5	6.9	3.8	0.0	3.0	.54
College degree	21.7	17.2	26.9	22.2	21.2	
Graduate school	74.8	75.9	69.2	77.8	75.8	
Mullen VR T (mean, *SD*)	52 (13.2)	55 (10.1)	52.9 (15.3)	55.4 (11.9)	45.9 (13.4)	.12
SABS T1, T2 weighted total (mean, *SD*)	31.3 (12)	32.2 (11.1)	31 (13.6)	34.6 (10.5)	27.9 (12.2)	.76
ROWPVT-4 age equivalency (mean, *SD*)	36.4 (15.6)	39 (12.1)	38.6 (18.5)	38.6 (13.2)	30.6 (16.9)	.11
Child behavior during book sharing[a] (mean, *SD*)	9.2 (3.6)	10.8 (2.8)	9 (4)	10.1 (2.7)	7.1 (3.7)	.02
EA-CP vocabulary test total (mean, *SD*)	8.6 (5.4)	9.1 (4.4)	9 (6.2)	10.2 (4.8)	6.5 (5.6)	.37

Note. EA-CP = Early Achievements for Child Care Providers; BAU = business-as-usual; Mullen VR = Mullen scales of early learning visual reception subscale; SABS = social communication assessment in book sharing; ROWPVT-4 = Receptive One-Word Picture Vocabulary Test-4.

[a]Provider rating of child behavior during book sharing.

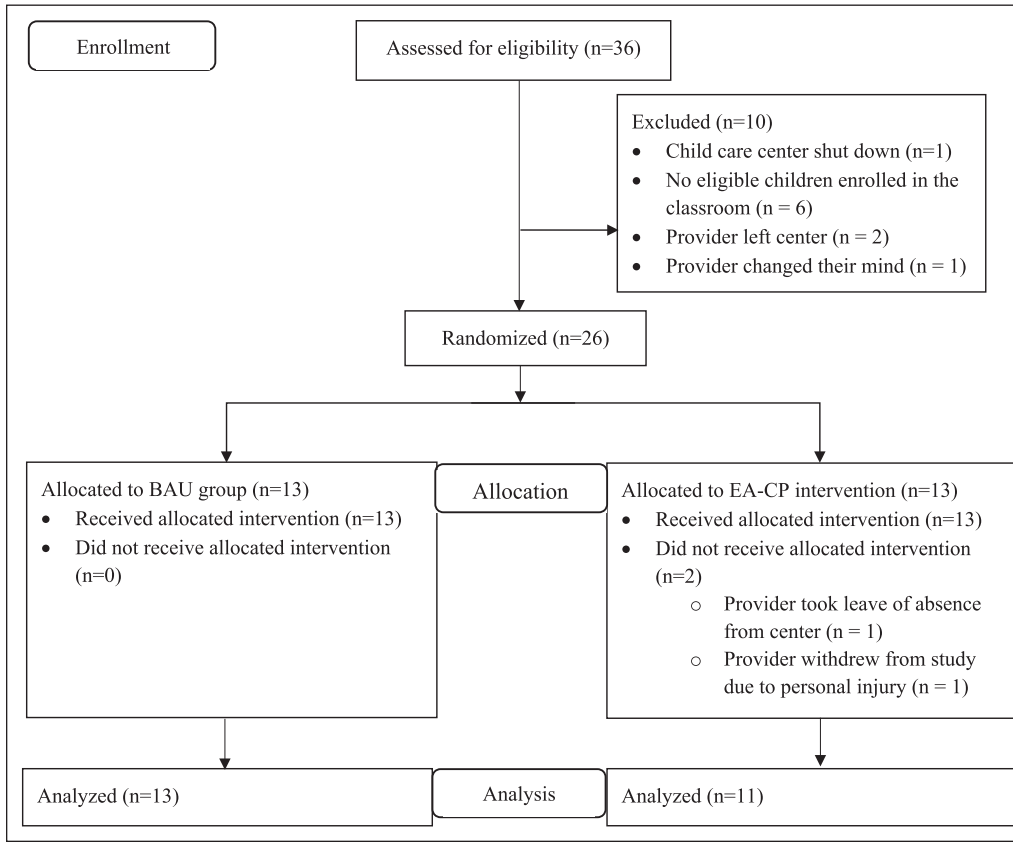


Figure 1. Provider Recruitment and Randomization Flow Diagram, From Consent Through Data Analysis.
Note. BAU = business-as-usual; EA-CP = Early Achievements for Child Care Providers.

workforce attrition (Harry et al., 2022). These adaptations included an abbreviated implementation period (12 weeks instead of the original 24 weeks); omission of EA-CP goals targeting joint attention and motor imitation; and elimination of two instructional implementation contexts (retaining only the book-sharing context).

Shared book reading offered a familiar context for providers' implementation because they already conducted nearly daily read-alouds; this minimized the amount of new learning required and the need to modify the classroom routine. Furthermore, providers' use of storybooks offered them a playbook to support consistent, high-dosage provision of story-embedded antecedents to engage children's language processing and embodiment of story-related concepts and event sequences (Feuerstein & Landa, 2020; Toub et al., 2018). For children, the story book provided a shared knowledge context and visual supports (illustrations, props) to scaffold language and communicative exchanges.

Key ingredients of the EA-CP book-sharing instructional approach included: (a) targeting receptive and expressive vocabulary, story-related gesture imitation, and story-related communication initiation and responsiveness within peer-to-peer engagement opportunities to promote children's linguistic formulation for authentic communicative purposes; (b) repeated (during 2-week intervals per book) use of strategically selected children's picture books having a simple and clear plot with non-abstract illustrations depicting simple scenes and affording repeated

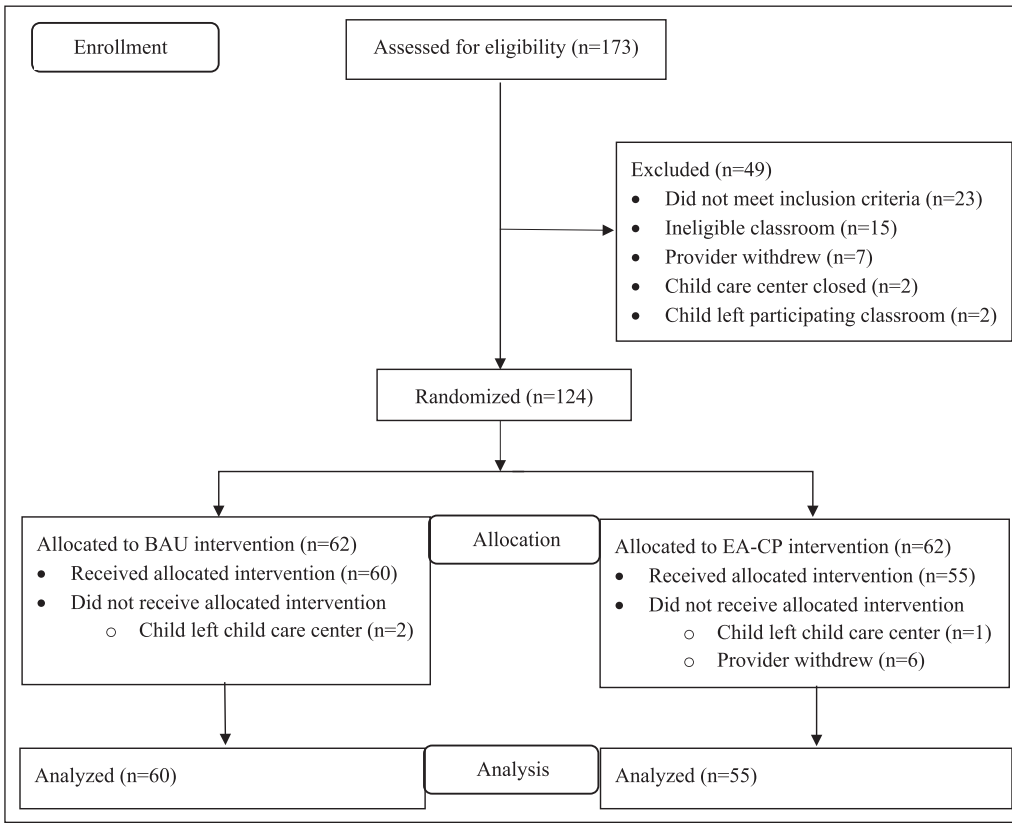


Figure 2. Child Recruitment and Randomization Flow Diagram From Consent Through Data Analysis.
Note. BAU = business-as-usual; EA-CP = Early Achievements for Child Care Providers.

opportunities to target a set of five vocabulary words; (c) implementing target-related activities-per-page-spread using props that would support children's learning of vocabulary semantic features, formation of a semantic network related to target vocabulary, and readily elicit children's expressive use of target vocabulary; and (d) using selected NDBI strategies. NDBI strategies included (a) providing clear antecedent cues, (b) prompting (e.g., modeling and pausing), (c) rewarding child attempts to respond to the antecedent cue through contextually related (natural) reinforcement, and (d) expanding/extending child utterances to provide more complete semantic representations (McKeown, 2019). EA-CP features overlap with the dialogic reading approach in that both approaches emphasize child verbal participation. However, EA-CP emphasizes children's expressive language involving target vocabulary use and providers' systematic use of specific NDBI strategies that are not clearly defining characteristics of, but are not incompatible with, dialogic reading.

Overview of EA-CP PD

The EA-CP PD program consisted of two workshops and 12 coaching sessions, described below. Three primary innovations were incorporated into the EA-CP PD approach. First, the PD content focused on training providers to implement NDBI strategies embedded within whole-group instruction. Second, the PD provided relevant implementation guidance and videos of other providers' authentic implementation of NDBI strategies, allaying concerns by

some providers that NDBIs are not aligned with early childhood educators' pedagogy, but rather fall within the domain of clinicians (Pfeiffer et al., 2022). Thus, the PD spotlighted the targeting of child vocabulary and communication skills, with well-defined antecedent cues and prompt hierarchies associated with each targeted skill. PD procedures also involved training providers to motivate children's sustained engagement and verbal initiations through integrating child choice and the use of natural rewards. Third, providers were trained to use illustrations and text in children's books as cues for how to plan engaging activities, use props, and provide opportunities to deliver a high dosage of specific child language and communication antecedent cues. The primary investigator (first author) and two of the expert coaches (second and last author) selected four "mentor books" (e.g., *Roadwork* by Sally Sutton) with colorful illustrations, clear and relatable story plot, and age-appropriate target vocabulary to foster providers' learning of the EA-CP intervention elements. Five vocabulary words were chosen by the study team to target in each book that were likely to be novel to the children and were repeated several times throughout the book. Props were selected to accompany each book to permit child-embodied learning of target vocabulary. For example, the target vocabulary words for *Roadwork* were *dirt*, *hammer*, *road*, *roll*, and *truck.* The props provided for this book included kinetic sand, a toy hammer, road tape, art rollers, a toy truck, a shovel, and mini construction cones. Laminated pictures representing each of the five target vocabulary words were also provided for posting in their book-sharing area to encourage children's vocabulary generalization (i.e., the illustration of the word "hammer" in the book vs. a photo of an actual hammer) and to encourage children to initiate language (i.e., "I see another hammer!"). The first two books and accompanying props were delivered to EA-CP providers at their centers before the first workshop was received (both conditions). The third and fourth were delivered by the coaches during coaching sessions a week before the EA-CP providers were expected to use each one.

Workshops

Providers participated in two 3-hr virtual workshops hosted using Zoom technology. For EA-CP providers, the Day 1 and Day 2 workshops occurred prior to the first and after the second coaching sessions, respectively. A total of 12 EA-CP providers attended the Day 1 workshop together and a make-up workshop was held with one provider. The Day 2 workshop was hosted on three different days to accommodate providers' schedules. Six providers attended each of the first 2 days and one provider attended the last day. The BAU group completed workshops about 4 weeks ($M = 3.89$ weeks; $SD = 0.85$, range = 1.8–4.8) after the post-training, about 8 weeks before the maintenance fidelity measurement.

Workshops were facilitated by expert coaches (a master's level early intervention specialist and two speech-language pathologists). All three coaches facilitated the Day 1 workshop and two of the coaches (an early intervention specialist and a speech-language pathologist) facilitated the Day 2 workshop. Coaches delivered didactic, lecture-based instruction using PowerPoint presentations with video models of previously trained providers' EA-CP implementation (Coogle et al., 2022). Five and seven video models were shown during the Day 1 and 2 workshops, respectively. Consistent with adult learning principles, a supportive community of learners was created by integrating *active learning opportunities* into didactic elements of the PD including observation, feedback, collaboration, and discussion (Knowles et al., 2015). For example, after the coaches showed the video models of the instructional strategies and/or demonstrated them, the providers selected pages in the book being used in the workshop and planned how they would integrate the instructional strategies and props during shared book readings. Then participants discussed their ideas with the larger group and received feedback from the coaches and other providers. Afterward,

providers went into virtual breakout rooms once during each workshop to *practice role-playing* the instructional strategies using study-provided books (with targeted vocabulary) and props in small groups. This enabled providers to receive feedback in a *respectful learning environment* within a community of peers.

A trained study staff member assessed workshop fidelity of the participating coaches' live workshop delivery during the first presentation of the Day 1 and Day 2 workshops for the EA-CP providers using a 15-item *Workshop Implementation Form* examining adherence and quality variables. The Adherence Domain included the following items which were rated on a 0 (did not complete) or 1 (completed) scale: (a) began workshop on time; (b) provided breaks for attendees; (c) delivered all slides; (d) narrated video exemplars of EA-CP instructional strategies; (e) facilitated group discussion; (f) led breakout sessions; (g) engaged attendees in a role play of EA-CP instructional strategies; and (h) modeled EA-CP instructional strategies when appropriate. The Quality Domain consisted of the following items: (a) provided explicit feedback with concrete examples; (b) phrased feedback in a positive, encouraging way; (c) elicited providers' self-reflection during breakout sessions; (d) attempted to understand attendees' unique classroom situations; (e) demonstrated a demeanor of respect for attendees' opinions, perspectives, and insights; (f) provided supportive feedback during breakout sessions and role-play activities; and (g) provided constructive feedback during breakout sessions and role-play activities. Adherence ($M = 94\%$, $SD = 8.49\%$) and quality ($M = 96.5\%$, $SD = 4.95\%$) were high across workshops.

Coaching

All providers in the EA-CP group received 12 coaching sessions in their classrooms across the PD period ($M = 16.16$ weeks, $SD = 1.85$, range = 14.2–19.4 weeks). These synchronous 1:1 coaching sessions focused on providers' book-sharing activity. Practice-based coaching (PBC; Snyder et al., 2022) was used. Each coaching session included three components. First, coaches conducted a focused observation during providers' book-sharing activity, modeling or prompting as appropriate to support the implementation of EA-CP and NDBI strategies. Next, coaches conducted a debriefing, guiding providers' reflection about their implementation of the instructional strategies and their progress toward goals defined in the prior week's collaborative action plan. Supportive and constructive feedback was offered. Finally, the provider and coach co-constructed up to three priority goals to guide the provider's systematic intervention implementation and jointly defined an action plan to achieve those goals during the next week. A written summary of the goals/action plans was emailed to providers after each coaching session to ensure shared reference for review at the next coaching session. Coaches and providers jointly identified resources that the providers desired for addressing the action items (e.g., cue cards to remind them to give children more opportunities to communicate about targeted vocabulary words) and how to access them (e.g., by making, purchasing, or borrowing them) (Snyder et al., 2022). Providers' action plan goals targeted implementation of specific instructional strategies and/or engagement of students during a book-sharing activity. Duration of coaching sessions, including all three PBC components, was about 30 min.

Two trained study staff members rated 30% of randomly selected coaching audio tapes using a 21-item coaching fidelity protocol addressing adherence (e.g., gives explicit examples in feedback about at least two child learning opportunities offered by the provider) ($M = 95.93\%$, $SD = 7.2\%$) and quality (e.g., acknowledges provider's strengths and discusses how these can help the provider implement effective teaching practices) ($M = 97.92\%$, $SD = 4.29\%$); coaching protocols were implemented as intended.

Provider Outcome Measures

Provider Implementation Fidelity. All providers in both conditions were videorecorded in their classrooms by trained research assistants blinded to group membership while conducting whole-group book reading at baseline, mid-training (after the EA-CP group's sixth coaching sessions), post-training, and follow-up (maintenance). Post-training videorecording occurred for the EA-CP group following their final coaching session ($M = 4.36$ days, $SD = 1.86$, range = 2–8 elapsed days), and for the BAU group 16 weeks after PD onset ($M = 16.03$ weeks, $SD = 0.72$, range = 15.2–17.4 elapsed weeks from pre to post). The maintenance videorecording for both groups occurred 12 weeks after post-training videorecording ($M = 12.50$ weeks, $SD = 0.51$, range = 12–13.4 elapsed weeks).

Providers' implementation of the targeted evidence-based instructional practices was examined using an adaptation of the *EA-CP Book Sharing Implementation Fidelity Form* and coding manual (Feuerstein & Landa, 2020). The adapted form consisted of four sections with 29 total items: (a) Environmental Arrangement (2 items); (b) Structuring the Book Sharing Activity (5 items); (c) Strategy Implementation (NDBI, providing opportunities for the instructional targets) (17 items); (d) Overall Quality (5 items). Each item was rated on a 3-point Likert-type scale (0 = *no implementation*; 1 = *attempted but did not meet criteria for full and correct implementation*; 2 = *full and correct implementation*). Percent fidelity was calculated as the total sum of points assigned to each item divided by the number of total points possible (58) × 100.

Inter-rater reliability was conducted by two independent, trained coders, blinded to group membership and data point on 33.7% (32/95 videos) across providers' pre-training, mid-training, post-training, and maintenance videos. The intraclass correlation coefficient (ICC; Koo & Li, 2016) was calculated at each of these timepoints and ranged from .93 to .98 (average .97 across timepoints), indicating excellent reliability.

Provider Knowledge Assessment. The Provider Knowledge Assessment was a study-developed measure to assess knowledge of the NDBI strategies, as well as child communication behavior, by identifying them in video clips. The assessment consisted of 19 multiple-choice questions (one point each) and one free-response item (two possible points), for a total of 21 possible points. Ten of the multiple-choice questions corresponded with a video clip of a child care provider from a previous cohort demonstrating one of the EA-CP instructional strategies during book sharing. The other questions elicited providers' knowledge of key concepts taught during the training such as identifying developmentally appropriate vocabulary words and the benefits of repeated book reading. Providers in both groups completed the assessment electronically over Zoom with study staff present (preventing using the internet to find answers) at baseline (before the first workshop) and post-training (after the EA-CP group's final coaching sessions). BAU providers completed the post-assessment before participating in the workshops.

Provider Self-Efficacy. A retrospective *Provider Self-Efficacy Scale* was administered to all providers after the EA-CP group completed the intervention. This 16-item scale was used to examine providers' perceptions of their implementation of instructional strategies taught in the EA-CP PD program. Providers completed two sets of ratings after the PD program ended: one for their retrospective perceived self-efficacy before the PD program started and one for their current perceived self-efficacy at the end of the PD program. A retrospective approach for the pre-training measurement was used because of the tendency for early childhood educators to overestimate their self-efficacy (Cunningham et al., 2009). Sample items included: (a) creating opportunities for children to imitate adults and peers, (b) baiting the classroom environment with opportunities for children to say target vocabulary words, and (c) responding to children's communicative

attempts. Each item was rated on a 3-point Likert-type scale: 1 = *not at all confident*, 2 = *somewhat confident*, and 3 = *highly confident*. A total score was calculated by summing all individual item scores (48 maximum points possible).

Child Outcome Measures

Assessments by study staff were administered one-on-one by a trained study staff member with each child in their child care centers in a space away from classroom activities.

Social Communication Assessment in Book Sharing (SABS). The SABS (Feuerstein & Landa, 2020) was administered to children at pre- and post-training to assess for changes in specific social communication behaviors. During the assessment, the examiner engaged the child in a shared story book reading activity while administering five structured probes per each of six behaviors (initiation of joint attention, response to joint attention, motor imitation, expressive vocabulary, receptive vocabulary, and following directions in the use of book-related object play using props). The examiner administered these probes at standardized pre-determined points in the story, pausing after each one to allow the child to initiate or respond. Each probe was administered up to two times. If the child answered correctly, the examiner continued with the assessment. If the child answered incorrectly, the examiner readministered the item. Structured probes for initiation of joint attention, receptive vocabulary, and following directions were readministered verbatim. Probes for response to joint attention and expressive language were readministered with a prompt ("Look!" and "What is this?," respectively). Probes for motor imitation were only readministered with the prompt "Watch me!" if the child was not paying attention during the first trial. The assessment was videorecorded for later coding by three trained research assistants blinded to child group and data point. Responses during both trials were scored as correct or incorrect. Correct responses during the first administration were scored as two points, while correct responses during the second trial were scored as one point (maximum possible score = 60).

Provider Rating of Child Behavior During Book Sharing. The *Provider Rating of Child Behavior During Book Sharing* is a study-developed measure adapted from the Teacher Rating of Oral Language and Literacy (Dickinson et al., 2003). Providers completed this brief seven-item measure for each of the child participants in their classroom at pre- and post-training. It was designed to capture children's engagement and enjoyment of shared book reading as well as their use of book-related vocabulary and concepts during the school day outside of shared book reading. Sample items included: (a) How often does the child join story time when stories are read in a group? (b) How often does the child say words during non-book-sharing activities during the school day that they heard during story times? and (c) How often does the child comment or ask questions related to the story during story time? Providers rated each item using a 3-point Likert-type scale (1 = *rarely or never, one day a week or less*; 2 = *sometimes, some days in the week*; 3 = *often, nearly every day*). A total score was calculated by summing each item score, for a maximum of 21 possible points.

Receptive One-Word Picture Vocabulary Test-4 (ROWPVT-4). The ROWPVT-4 (Martin & Brownell, 2011), normed for ages 2 years and older, was administered as a standardized, norm-referenced measure of receptive vocabulary. The measure involves presenting children with a four-way discrimination task wherein they are to point to which of the four two-dimensional pictured objects, actions, or concepts matches the word spoken by the examiner. The ROWPVT-4 has a high degree of reliability for internal consistency, ranging from .94 to .98, with a median of .97 across all ages. Test–retest correlation for raw scores is .97 and .91 for standard scores.

Expressive Vocabulary Assessment. The *EA-CP Vocabulary Test* is a proximal, study-developed, 20-item expressive vocabulary measure consisting of 20 cards picturing five targeted words from each of the four books used in the EA-CP PD program. A total of 18 nouns were selected for the assessment and two verbs that could be gestured; most words would be classified as Tier 1 (e.g., dough, nest, yawn), some were Tier 2 words (e.g., astronaut). The examiner showed each picture to the child individually and asked, "What is this?" for each of the nouns and "What are they doing?" for the verbs. Children received a score of 1 if they correctly labeled the item or 0 if their response was incorrect, for a total of 20 possible points. A second trial was delivered if the child's response was unintelligible, if the child named the wrong part of the pictured object, or if the child's response was too specific (e.g., tulip for flower); correct responses were assigned a score of 1.

Analysis

The lmerTest package (Kuznetsova et al., 2015) in R (R Core Team, 2020) was used to perform mixed effects linear regression models. For both provider and child regression models, the primary independent variable was an interaction term between the condition (EACP vs. BAU) and timepoint (post vs. pre) and the model included a random effect for each individual (provider or child) to account for repeated observations over time. Provider regression models also adjusted for the frequency of students with DD in the classroom. Child outcome models did not account for child nesting within each classroom due to the relatively small number of students within some of the classrooms (range 2–9, mean 4.8, median 4), which caused issues with model convergence. For child regression models, analyses were performed in the total sample as well as in the stratified non-delayed and DD samples. These regression models produced a beta estimate, 95% confidence interval, and *p*-value. The criterion for statistical significance was set at $p \leq .05$. Cohen's *d* estimates, using the Hedge's *g* correction and the pooled standard deviation, were also calculated as a measure of unadjusted effect size. The magnitude of these effect sizes was assessed according to the thresholds provided by Cohen, 1992, that is, $|d| < 0.2$ "negligible," $|d| < 0.5$ "small," $|d| < 0.8$ "medium," otherwise "large."

Results

Descriptive

Demographic and developmental characteristics of child participants are reported in Table 1, stratified by the presence of child developmental status and PD condition. Examination of baseline characteristics revealed no significant differences across any of the variables between EA-CP and BAU groups, with the exception of the *Provider Rating of Child Behavior During Book Sharing*, which was significantly higher in the EA-CP group ($t = -2.29$; $p = .02$).

Providers in the EA-CP group reported the number of their book-sharing administrations and attendance of participating children during these administrations each week. Providers reported an average of 32.36 readings ($SD = 7.80$, range: 16–38) during the PD period. Participating children attended an average of 28.63 readings ($SD = 6.99$, range: 13–38).

Provider Outcomes

At pre-training, providers in both conditions were implementing evidence-based instructional strategies about 25% of the time: EA-CP $M = 25.86$ ($SD = 15.73$); BAU $M = 28.25$ ($SD = 10.16$). By mid-training (coaching session 6/12), the EA-CP providers had attained 77.12% ($SD = 15.40$) fidelity in the implementation of NDBI strategies, and they retained that level of fidelity through the end of the training period ($SD = 23.97$). At maintenance, EA-CP

Table 2. Mixed Effects Models: Effect of the PD Program on Provider Outcomes.

Outcome	Predictor	Estimate	95% CI	*p*	*g*	Effect size
Book fidelity overall percentage	EA-CP	–2.94	[–14.62, 8.74]	.62	—	—
Book fidelity overall percentage	Timepoint: Mid	0.00	[–8.44, 8.44]	1.00	—	—
Book fidelity overall percentage	Timepoint: Post	1.99	[–6.45, 10.43]	.64	—	—
Book fidelity overall percentage	Timepoint: Maint[a]	12.75	[4.1, 21.39]	<.001	—	—
Book fidelity overall percentage	Frequency of DD	−5.82	[−23.92, 12.28]	.51	—	—
Book fidelity overall percentage	**EA-CP × Mid**	**51.25**	**[38.79, 63.71]**	**<.001**	**4.19**	**Large**
Book fidelity overall percentage	**EA-CP × Post**	**49.26**	**[36.80, 61.72]**	**<.001**	**2.67**	**Large**
Book fidelity overall percentage	**EA-CP × Maint[a]**	**33.96**	**[21.36, 46.57]**	**<.001**	**2.12**	**Large**
Knowledge assessment	EA-CP	0.47	[−1.91, 2.86]	.69	0.73	Medium
Knowledge assessment	Timepoint: Post	1.08	[−0.38, 2.54]	.14		
Knowledge assessment	Frequency of DD	−2.07	[−6.37, 2.23]	.33		
Knowledge assessment	EA-CP × Post	1.83	[−0.33, 3.99]	.09		
Self-efficacy total	EA-CP	3.09	[−2.07, 8.25]	.23	0.38	Small
Self-efficacy total	Timepoint: Post	8.75	[5.04, 12.46]	<.001		
Self-efficacy total	Frequency of DD	0.14	[−9.33, 9.6]	.98		
Self-efficacy total	EA-CP × Post	2.34	[−3.02, 7.7]	.37		

Note. Model includes 24 providers (13 BAU, 11 EA-CP). Bolded text shows statistically significant group by time interaction effects. PD = professional development; CI = confidence interval; EA-CP = Early Achievements for Child Care Providers; DD = delay or disorder; BAU = business-as-usual.
[a]Maint = Maintenance datapoint.

providers demonstrated 72.57% (*SD* = 13.34) fidelity. In contrast, the BAU group exhibited no improvement in the mid- or post-training assessments. However, following their completion of the workshop between the post-training and maintenance fidelity measurement, the BAU group's fidelity increased from 30.24% (*SD* = 9.74) at post-training to 41.09% (*SD* = 13.49) at the maintenance measurement (see Table 2, and Supplemental Table 1-S and Figure 1-S).

Results of the linear mixed effects model revealed that the EA-CP providers demonstrated significantly higher fidelity gains at all timepoints compared to pre-training than did the BAU providers (e.g., at Post, β = 49.26; 95% CI 36.80, 71.72; $p < .001$; $g = 2.67$). The *Provider Knowledge Assessment* and *Provider Self-Efficacy Scale* total scores did not differ significantly from pre- to post-training between EA-CP and BAU groups (see Supplemental Table 1-S). However, group × time effect sizes were positive for the knowledge and self-efficacy measures (medium and small, respectively) favoring the EA-CP group (Table 2 and Supplemental Figure 1-S).

Child Outcomes

Assessments of children occurred at pre- and post-training. When children with and without delays were analyzed together (Table 3), there was a significant group difference (EA-CP vs. BAU) from pre- to post-training in the *Expressive Vocabulary* total score, favoring the EA-CP group (β = 1.58; 95% CI 0.43, 2.73; $p = .01$; $g = .51$). In addition, there was a significant group difference over time in the *Provider Rating of Child Behavior During Book Sharing* total score, favoring the EA-CP group (β = 1.23; 95% CI 0.03, 2.43; $p = .04$; $g = .38$). No significant group

Table 3. Mixed Effects Models: Effect of EA-CP Implementation on Vocabulary and Communication Outcomes Among All Children and Children With Delays.

Child population	Outcome	Predictor	β estimate	95% CI	*p*	*g*	Effect size
	Child Behavior During Book Sharing	EA-CP	1.51	[0.33, 2.70]	.01	.38	Small
All Children	Child Behavior During Book Sharing	Timepoint: Post	0.90	[0.07, 1.73]	.03		
	Child Behavior During Book Sharing	**EA-CP × Post**	**1.23**	**[0.03, 2.43]**	**.04**		
	Expressive Vocabulary Test Total	EA-CP	0.91	[−0.95, 2.76]	.34	.51	Medium
All Children	Expressive Vocabulary Test Total	Timepoint: Post	3.20	[2.41, 3.99]	<.001		
	Expressive Vocabulary Test Total	**EA-CP × Post**	**1.58**	**[0.43, 2.73]**	**.01**		
Children with Delays	Child Behavior During Book Sharing	EA-CP	1.88	[0.06, 3.70]	.04	.43	Small
	Child Behavior During Book Sharing	Timepoint: Post	1.52	[0.37, 2.66]	.01		
	Child Behavior During Book Sharing	EA-CP × Post	1.41	[−0.32, 3.14]	.11		
Children with Delays	EA-CP Vocabulary Test Total	EA-CP	2.48	[−0.36, 5.33]	.09	.46	Small
	EA-CP Vocabulary Test Total	Timepoint: Post	3.36	[2.27, 4.46]	<.001		
	EA-CP Vocabulary Test Total	EA-CP × Post	1.44	[−0.20, 3.09]	.08		

Note. Models with all children include 115 children total (60 BAU and 55 EA-CP). Models with children with delays only include 59 children total (33 BAU, 26 EA-CP). EA-CP = Early Achievements for Child Care Providers; CI = confidence interval; BAU = business-as-usual.

differences over time for SABS weighted total or ROWPVT-4 age equivalency scores were detected.

Among children with DD (Table 3), group × time effect sizes were positive, favoring the EA-CP group, for the *Expressive Vocabulary* total score and the *Provider Rating of Child Behavior During Book Sharing* total score, from pre- to post-training compared to the BAU group (*g*'s = .46, .43, respectively); estimates approximated but did not reach our threshold for significance (*p's* = .08, .11, respectively). No significant group differences over time were detected among children with delays in the EA-CP versus BAU group for the SABS weighted total score or ROWPVT-4 age equivalency score (see Supplemental Table 2-S). For results for children with no delays, see Supplemental Table 3-S.

Discussion

This pilot cluster RCT culminated from a multi-phase iterative process guided by implementation science principles to develop the PD program of focus. The PD development process paved the way for this RCT to address gaps in research examining the effects of PD to support providers' implementation of NDBI strategies during whole group, inclusive book sharing to advance vocabulary and communication learning in young children with and without DD. Despite abbreviating the PD and EA-CP instructional approaches, EA-CP-trained providers' gains in fidelity of implementation of evidence-based instructional practices were significantly greater than that of BAU providers and were sustained, without when selecting their own books and target

vocabulary after training ended. This is particularly notable due to the PD program's delivery during the COVID-19 pandemic. Significant positive effects of providers' EA-CP implementation on child outcomes (expressive vocabulary and story-related engagement and communication) were detected when children with and without delays were combined within groups; pre-post increases in the EA-CP group were significantly greater than in the BAU group. When comparing pre-post gains only in children with delays across both groups, results show promise of PD effects for improving targeted skills (g's $> .4$) but did not reach statistical significance. Results demonstrate the feasibility and effectiveness of the EA-CP PD program for empowering providers' NDBI implementation and generalization during book sharing to increase language learning in young children with and without DDs.

Impact of the COVID-19 Pandemic and Implications for Effects on Children With DD

This PD program was conducted during the COVID-19 pandemic, intensifying (Markowitz & Bassok, 2022) pre-pandemic challenges (Hamre, 2014) within early childhood education settings. Due to directors' concerns about compounding pandemic stressors, demands on providers were minimized by: (a) shortening the PD program to 12 weeks, and (b) reducing the workshop duration by half. Also, the EA-CP instructional approach was simplified by eliminating the implementation of motor imitation and joint attention goals, reducing the frequency of vocabulary production opportunities and of the richness and frequency of conversations involving vocabulary-related concepts (Farrant et al., 2011). EA-CP simplifications likely constrained detection of effects on the SABS, which includes items to assess joint attention and motor imitation. In addition, many providers had to pause their participation periodically due to center closures or their own illnesses. There also were extended child absences due to infection control policies prohibiting attendance of children exhibiting COVID-related symptoms or having a COVID-positive family member. These factors reduced dosage and consistency of language learning experiences afforded to children by EA-CP providers, likely constraining the detection of significance in comparisons involving just the TD or DD groups especially given small sample sizes. The DD group may have been particularly impacted, given their need for greater dosage than those without DD.

Importance of Multi-Faceted PD Design

At baseline, providers in both groups exhibited low fidelity of implementation of evidence-based instructional practices during whole-group book sharing. Yet providers in the EA-CP PD program, which was multi-faceted in design (e.g., interactive workshops, coaching, explicit implementation strategies, instructional targets, and use of mentor materials) demonstrated at least a three-fold gain in fidelity from pre- to post-training. Gain was detected by the sixth of 12 coaching sessions and sustained through the 12-week maintenance period. The low baseline fidelity of providers in both groups and the EA-CP providers' degree of gain aligns with our prior RCTs that employed a more complex version of EA-CP implemented with toddlers having DD (Feuerstein & Landa, 2020; Pfeiffer & Landa, 2022).

Two candidate explanations for the rapid increase in the EA-CP group's fidelity are likely. The first involves careful selection of study-provided books and props. Pre-selected books had the following qualities to support providers' rapid attainment of high-fidelity implementation: (a) simple stories that focused the provider's language on objects and events related to the target vocabulary rather than trying to explain complex plot twists and turns; (b) clear illustrations of objects and events that afforded reminders to encourage children to label, define, and discuss target vocabulary (related to dosage aspects of fidelity); and (c) text that offered rich opportunities to use the targeted vocabulary. These qualities enabled the books to serve as a playbook to

scaffold providers' ability to clearly identify on which pages they would present the antecedent cues, and for which goals, to encourage children's verbal initiations (Greenhoot et al., 2014). The second explanation pertains to implementation planning with coach-supported hands-on practice during the workshops. Practicing during the first workshop fostered providers' fluency with the intervention strategies prior to implementation in their classrooms. The coaches' presence in the workshops enabled providers to have questions addressed immediately and to receive feedback on missed opportunities and erroneous implementation. This experience likely bolstered their fidelity during their early in-class implementations.

Providers' sustained level of fidelity between the 12-week post-training and maintenance measurement interval indicates generalization to untrained instructional materials and is suggestive of adoption of NDBI strategies and the EA-CP instructional approach. The BAU group's 10% increase in fidelity between post-training and maintenance measurements may have been due to completing the workshops in that interval. Results indicate that coaching is needed to significantly effect practice change (Neuman & Cunningham, 2009).

Results of the knowledge assessment revealed that the EA-CP providers' gain from pre- to post-training was 2.69 times greater than that of BAU providers (medium effect size; not meeting the criterion for statistical significance). The discrepancy between training effects detected on the knowledge and self-efficacy measures in light of positive and significant effects on fidelity could be attributed to measurement factors (e.g., knowledge assessment too difficult), but highlights the importance of combining workshops with coaching on specific instructional practices to improve providers' practice of NDBI implementation. More research is needed to better understand the gap between EA-CP providers' modest post-training knowledge scores and their high implementation fidelity, and whether additional training to improve knowledge would have an additive effect on fidelity.

Provider NDBI Implementation in Book Sharing Improves Child Language Skills

Child gains exceeding what could be attributed to maturation were observed in whole-group analyses (children with and without delays) for the EA-CP group's expressive vocabulary learning and book-related engagement and communication outcomes. When examining only children with delays, positive effect sizes (small) were detected. While most children did not yet have a diagnosis of developmental language disorder, most were at elevated likelihood of receiving the diagnosis in later childhood based on baseline data (particularly ASQ data) and providers' concerns. Storkel and colleagues (2019) have shown that preschool-aged children with developmental language disorder (DLD) learn words during book sharing but forget newly learned words after the intervention ends. Since targeted vocabulary was assessed at pre- and post-training, children's memory of words taught months before the post-training measure could have faded. The PD effects on learning in children with DD in the present study are discrepant with our prior findings in inclusive child care settings where significantly greater gains were observed in the EA-CP group of autistic toddlers for social communication compared to a BAU group (Feuerstein & Landa, 2020). This is most likely due to reduced scope of intervention (attenuating SABS gains) and intervention consistency and dosage.

Limitations

One limitation is small sample size, likely limiting statistical power to detect significant group by time effects in children with DD. Small sample sizes are expected in pilot RCTs; results nonetheless demonstrated that the EA-CP PD program effectively improved providers' NDBI implementation during book sharing. Further, despite the limited number of children with delays in both

groups, the positive effect sizes on some child outcome measures indicate that the PD program shows promise for advancing language-related learning in young children with DD. Other limitations included the focus on a single instructional training context (but see Pfeiffer & Landa, 2022), providers' variability in targeting identified vocabulary during instruction, and variability in the frequency of providers' book-sharing implementation sessions, reducing the dosage.

Conclusions and Implications

The EA-CP PD program studied here had undergone a 3-year iterative development process, with the final iteration showing stakeholder confirmation of implementation feasibility, significant effects on early childhood education providers' fidelity, and significant effects on communication outcomes in children. This pilot RCT demonstrated that the EA-CP PD program empowered providers to implement, with high fidelity, NDBI strategies in the demanding and complex environment of child care, where their primary duty is to ensure the overall well-being of a full class of diverse children. Providers' sustained fidelity gains after training show promise for adoption of NDBI implementation. Theoretical implications for future PD research involve the need to adopt an implementation science-informed approach, particularly considering inner and outer contexts. Information gained would guide the preparatory work needed prior to PD implementation in early childhood education settings including provision of pre-PD support to equip directors to support providers' implementation (Pfeiffer et al., 2022), and assessment of providers' readiness for innovation and consistent intervention implementation. Practical implications include providing bootstrapping resources to enable providers to implement the intervention consistently and feasibly (e.g., providing mentor books and props early in the PD program to entrain strategy use), and expanding implementation to infuse explicit vocabulary instruction across contexts and over time to improve children's retention of learning (Marulis & Neuman, 2010). PD programs should provide frequent, distributed, embodied learning and practice opportunities with ongoing, medium- to long-term generalization learning opportunities to consolidate providers' NDBI implementation learning and children's expressive language learning. Policy implications include the need to invest in science-informed training of sufficient duration for early childhood education providers that combine didactic, guided practice and planning, and job-embedded, collaborative, and individualized reflective coaching.

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Supplemental Material

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